

DEVELOPMENT OF KLAES SURVEY (DEPARTMENT) MODULE WITH COMPENSATION MANAGEMENT AND AUTOMATED OCCUPANCY PERMIT WORKFLOW

CONCEPT NOTE

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1.0 BACKGROUND AND JUSTIFICATION

The Survey Department serves as the foundational technical unit within the Ministry of Land and Physical Planning, providing the spatial and measurement data that underpins all land administration activities. Currently, the department operates through multiple sections—Compensation, GKN (Government Known Numbers), Misc KN, LPKN (Layout Plan Known Numbers), Examination/Verification, and GIS Unit—each handling critical but largely manual processes.

The absence of an integrated digital system for survey operations has resulted in:

- Fragmented record-keeping across sections
- Manual field data collection for compensation exercises
- Delays in Occupancy Permit (OP) generation and processing
- Limited traceability of compensation beneficiaries
- Inefficient inter-departmental workflows between Survey, GIS, Deeds, and Land departments

The proposed KLAES Survey Department Module addresses these challenges by digitizing all survey operations, introducing mobile field data capture for compensation exercises, and establishing an automated workflow for Occupancy Permit generation that connects **Survey** → **GIS/KANGIS** → **Honourable Commissioner** → **Deeds** → **Land/Land OSS**.

2.0 MODULE OBJECTIVES

1. Digitize all survey sections (Compensation, GKN, Misc KN, LPKN, Examination/Verification, GIS)
2. Introduce mobile field data capture for compensation exercises with real-time farmer communication
3. Automate the Occupancy Permit workflow from generation to registration
4. Establish 50:50 land-for-land compensation formula automation
5. Enable economic tree valuation for monetary compensation
6. Integrate with existing KLAES modules (Deeds, Land, OSS, Digital Archive)

7. Provide real-time tracking of survey files and compensation cases

3.0 SCOPE OF WORK

3.1 Compensation Section Module

Web Interface:

- Compensation case registration and tracking
- Economic tree valuation calculator
- Land-for-land compensation calculator (50:50 formula)
- Compensation plan generation
- Summary reports (total area, plots for farmers, plots for government, land value, total fees)
- Beneficiary database (phone numbers, NIN, contact details)
- OP generation for compensation files

Mobile App (Field Data Capture):

- GPS-based farmland perimeter/area measurement
- Individual farm plot mapping within total farmland
- Economic tree counting and valuation
- Farmer biometric data capture (NIN, phone number)
- Photo documentation of farmland and economic trees
- Offline data capture with sync capability
- Real-time communication with farmers via SMS

Key Features:

- Monetary Compensation: Automated calculation of cash equivalent for economic trees
- Land for Land: Automated 50:50 plot allocation (e.g., 10 plots carved out → 5 plots to farmer, 5 plots to government for allotment)
- OP Generation: Automated creation of Occupancy Permits for compensation beneficiaries

3.2 GKN Section Module

- Indexing of existing GKN files (Government Known Numbers)
- Commissioning of new GKN files
- Government land inventory (Schools, Dams, Roads, Hospitals, Police Stations, etc.)
- Spatial data integration
- File tracking and status updates

3.3 Misc KN Section Module

- Indexing of existing Misc KN files
- Commissioning of new Misc KN files
- General inspection request management
- Applicant tracking and communication
- Inspection report generation

3.4 LPKN Section Module

- Indexing of existing LPKN files (Layout Plan Known Numbers)
- Commissioning of new LPKN files
- Layout plan number issuance
- Layout plan design management
- Instructions to Survey (I to S) generation
- List of Coordinates form management
- Field observation computation
- Plan validation and approval workflow

3.5 Examination/Verification Section Module

- Verification of compensation claims
- Cross-checking of survey data
- Dispute resolution tracking
- Examination report generation
- Integration with Compensation, GKN, Misc KN, and LPKN sections

3.6 GIS Unit Module

- Spatial data management and validation
- Parcel boundary verification
- Cadastral framework integration
- GIS certification of survey plans
- Coordinate system management
- Map production and printing (integration with KANGIS GIS)

4.0 AUTOMATED OCCUPANCY PERMIT WORKFLOW (THE "QUICK WIN")

The module introduces a seamless, automated workflow for Occupancy Permit generation and processing:

OP WORKFLOW CHAIN

1. SURVEY DEPARTMENT (Ministry)

- ↓ Generation of OP
- ↓ Compensation files or Government acquisitions
- ↓ Technical data capture and validation

2. GIS UNIT (KANGIS)

- ↓ Spatial certification
- ↓ Production/Printing of OP
- ↓ Quality assurance

3. HONOURABLE COMMISSIONER

- ↓ Electronic signature workflow
- ↓ Approval and authorization

4. DEEDS DEPARTMENT

- ↓ Electronic instrument registration
- ↓ Registration Details Sheet (RDS) generation
- ↓ Confirmation of Registration (CoR)

5a. LAND DEPARTMENT

- ↓ Further processing
- ↓ Opening of Land File
- ↓ Conversion to RofO/CofO
- ↓ (OPs have temp FileNo)

5b. LAND OSS

- ↓ Change of Ownership
- ↓ Transfer of Title (ToT)
- ↓ Fetch Existing File Record (FEFR)
- ↓ OP Subdivision/Merger handling

OP-to-File Linkage Scenarios (Handled by OSS FEFR Interface):

1. 1 OP → 2 Different Files (OP Subdivision): 1 OP, 2 Transfer of Titles
2. 2 OPs → 1 File (OP Merger): 2 OPs, 1 Transfer of Title
3. 1 OP → 1 ToT: Normal transfer
4. 1 OP → 2 ToTs: Original holder sells to Person A, who sells to Person B (current holder)

5.0 INTEGRATION POINTS

- KLAES Digital Archive: All survey documents indexed and archived
- KLAES Deeds Module: OP registration and instrument processing
- KLAES Land Module: File commissioning and further processing
- KLAES OSS Module: Change of ownership and FEFR interface
- KANGIS GIS: Spatial certification and OP production
- KLAES File Tracking: Real-time tracking of survey files and OPs
- KLAES CRM: Farmer and applicant communication management

6.0 TRAINING AND ONBOARDING

Target Groups:

1. Survey Department Staff (all sections)
2. Field Officers (Compensation Unit)
3. GIS Unit Staff
4. Examination/Verification Officers
5. Department Heads and Supervisors

Training Components:

- Web Module Training: 2 days (classroom + hands-on)
- Mobile App Training: 2 days (field practice)
- OP Workflow Training****: 2 days (cross-departmental) -DONE
- Administrator Training: 1 day (system configuration)
- Total Training Duration: 5 days per cohort

Onboarding Support:

- 3 months post-deployment technical support
- Dedicated support for Survey Department
- Weekly progress review meetings
- System optimization based on user feedback

7.0 EXPECTED OUTCOMES

1. 100% digitization of survey department operations
2. 70% reduction in OP generation and processing time
3. Real-time tracking of all compensation cases and survey files
4. Elimination of manual field data collection for compensation
5. Direct communication with compensation beneficiaries (SMS/Phone)
6. Automated 50:50 land allocation calculations

7. Seamless inter-departmental workflow (Survey → GIS → Commissioner → Deeds → Land)
8. Improved transparency and reduced fraud in compensation exercises
9. Comprehensive database of government land holdings (GKN)
10. Enhanced revenue generation through efficient OP processing and registration

8.0 IMPLEMENTATION TIMELINE

SN	Phase	Activity	Duration
1	Phase 1	Requirements finalization & system design	1 week
2	Phase 2	Module development (web + mobile)	2 weeks
3	Phase 3	Integration testing (KLAES ecosystem)	
4	Phase 4	User Acceptance Testing (UAT)	1 week
5	Phase 5	Staff training and onboarding	
6	Phase 6	Pilot deployment (Compensation Section)	
7	Phase 7	Full deployment (all sections)	
8	Phase 8	Post-deployment support & optimization	12 weeks
	TOTAL		4 weeks (~12 months)

NB: Implementation phases will run concurrently

9.0 RISK MITIGATION

SN	Risk	Mitigation Strategy
1	Resistance to change from manual processes	Comprehensive change management program, stakeholder engagement
2	Limited mobile network in rural areas	Offline-first mobile app design with sync capability
3	Data privacy concerns (farmer NIN, phone numbers)	NDPR compliance, encryption, access controls
4	Inter-departmental coordination challenges	Clear workflow documentation, cross-training, regular coordination meetings
5	Technical capacity gaps	Intensive training program, dedicated support team, user-friendly interface design

10.0 CONCLUSION

The KLAES Survey Department Module represents a critical enhancement to the Ministry's digital transformation agenda. By digitizing survey operations, introducing mobile field data capture, and automating the Occupancy Permit workflow, the module will:

- Improve service delivery to compensation beneficiaries and land applicants
- Enhance operational efficiency across Survey, GIS, Deeds, and Land departments
- Increase transparency and reduce opportunities for fraud
- Generate accurate, real-time data for decision-making
- Strengthen the Ministry's revenue base through efficient OP processing

The module aligns perfectly with the State's digital governance objectives and will position Kano State as a leader in innovative land administration practices in Nigeria.

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